

Total No. of Questions : 10]

SEAT No. :

P3733

[5561]-130

[Total No. of Pages : 5

B.E. (Mechanical Engineering)
REFRIGERATION AND AIR-CONDITIONING
(2012 Pattern) (Semester - I) (Theory) (402041)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer five questions from the following.
- 2) Draw neat labelled diagrams; wherever necessary.
- 3) Use of electronic calculator is permitted.
- 4) Use of Steam Table, Psychrometric chart, Refrigerant Property Table is Permitted.
- 5) Use of Programmable calculator is not allowed.
- 6) Assume suitable/standard data, if necessary.

Q1) a) Enumerate the limitations of Reversed Carnot Cycle? **[4]**

- b) A refrigerating plant; operating on Reverse Carnot Cycle, is required to produce 2.5 Tonnes of ice per day at -4°C from water at 20°C . If the temperature range in the compressor is between 25°C and -6°C , Calculate the power required to drive the compressor.

Assume Latent heat of ice 335 kJ/kg . ($L_{\text{ice}} = 335 \text{ kJ/kg}$); Specific heat of ice 2.1 kJ/kg K ($C_{p\text{-ice}} = 2.1 \text{ kJ/kg K}$); Specific heat of water 4.187 kJ/kg K ($C_{p\text{-water}} = 4.187 \text{ kJ/kg K}$). **[6]**

OR

Q2) a) Complete the table with proper Refrigerant Designation/Chemical formula? **[4]**

Sr. No.	Refrigerant	Chemical Formula	Designation
Inorganic Refrigerant			
01	Water	H_2O	-----?
02	Carbon Dioxide	CO_2	-----?
Organic Refrigerant			
03	Difluoro-Dichloro Methane	CCl_2F_2	-----?
Hydro carbons			
04	Propane	C_3H_8	-----?

- b) A Bell-Coleman refrigeration cycle operates between pressures of 4 bar and 16 bar. The air temperature after heat rejection to surrounding is 37°C and air temperature at the exit of refrigerator is 7°C . The isentropic efficiencies of turbine and compressor are 0.85 and 0.80 respectively. Determine Compressor and Turbine work per ton of Refrigeration (TR) and COP of the system? Assume $\gamma_{\text{air}} = 1.4$ and $C_{p\text{-air}} = 1.005 \text{ kJ/kg K}$. **[6]**

P.T.O.

- Q3) a)** Explain the effect of Sub-cooling and Superheating of refrigerant on COP with the help of T-S and P-H diagram in Vapour Compression Refrigeration System? [6]
- b)** Explain the effect of CFC's on Ozone depletion and Global warming? List out the two alternative refrigerants to CFC-12 in domestic refrigerators? [4]

OR

- Q4) a)** In an Aqua-Ammonia vapour absorption refrigeration system; the heat is supplied to NH_3 generator by condensing steam at 2 bar and 98% dry. The temperature in the refrigerator is to be maintained at -5°C , when the surrounding temperature is at 30°C . Find the maximum possible COP of the system, if the saturation temperature of steam corresponding to 2 bar is 120°C and latent heat of steam is 2201 kJ/kg.

If the refrigeration load is 20 TR and actual COP is 75% of the maximum COP, find the mass of steam required in kg/hr. Assume, only latent heat of steam is utilised for heating purpose? [6]

- b)** What are the advantages and disadvantages of Vapour absorption refrigeration system over Vapour compression refrigeration system? [4]

- Q5) a)** Does "Air-conditioning systems" are required in surgery rooms of Hospitals. Justify your answer? [4]
- b)** State and explain the factors which affect Comfort of a Human body? [4]
- c)** Atmospheric air at 0.965 bar enters the adiabatic saturator. The wet and dry bulb temperature is 20°C and 31°C respectively during adiabatic saturation process. [8]

Determine:

- Humidity ratio of entering air,
- Vapour pressure
- Relative Humidity.

Temperature in $^\circ\text{C}$	Saturation Pressure in bar	Enthalpy of Saturation Vapour in kJ/kg
20	0.02337	2538.2
31	0.04491	2558.2

OR

- Q6) a)** Define the following terms: [4]
- Grand sensible heat factor (GSHF)
 - Infiltration and Ventilation air
 - Room sensible heat factor (RSHF)
 - Outside air total heat load (OATHL)
- b) Explain the physical significance of bypass factor with reference to a heating coil? [4]
- c) A corporate bank located in Pune city at 30°N latitude has the following loads. [8]
- | | |
|--------------------|----------|
| Room sensible heat | 58.15 kW |
| Room latent heat | 14.54 kW |
- The summer outside and inside design conditions are:
- | | |
|---------|-----------------------|
| Outside | 40°C DBT and 27°C WBT |
| Inside | 25°C DBT and 50% RH |
- 70 cmm of ventilation air is used.
- Determine the following:
- Ventilation load
 - Grand total heat load
 - Effective sensible heat factor
- Assume a Bypass factor of cooling coil as 0.15; humid specific heat as 1.022 kJ/kg K. ($C_{pm} = 1.022$ kJ/kg K) and density of air as 1.2 kg/m³ ($\rho_{air} = 1.2$ kg/m³).
- Q7) a)** Explain with neat Sketch any Two of the following Air-conditioning system: [6]
- Unitary air-conditioning system
 - Industrial air-conditioning system
 - Packaged air-conditioning system
- b) Write a note on “operating and safety controls” of refrigeration and air-conditioning systems. List out minimum 08 safety controls of the system? [6]
- c) Why high and low pressure cut-outs are provided for refrigerant compressors? [4]

OR

- Q8)** a) What do you mean by Evaporators? Explain the functioning of flooded evaporators with the help of neat sketch? [6]
- b) Explain the working of Thermostatic Expansion valve with the help of neat sketch? Enumerate its advantages and disadvantages over other type of expansion devices. [6]
- c) Write a note on capacity control of reciprocating compressors? [4]

- Q9)** a) Explain any two of the following air distribution systems: [6]
- i) Ejector system
 - ii) Downward system
 - iii) Upward system
- b) A duct of rectangular cross section 600 mm x 400 mm carries 90 m³/min of air having density of 1.2 kg/m³. Determine the equivalent diameter of a circular duct when; [6]
- i) When the quantity of air carried in both the cases is same; and
 - ii) When the velocity of air in both the cases is the same.
- If friction factor (f) = 0.011, find the pressure loss per 100 m length of the duct?
- c) Write a note on “Duct Design”. Explain any one of the methods in details? [6]

OR

- Q10)** a) Explain in details about the function of “Air Handling Unit (AHU’s)” in an Air conditioning application? [6]
- b) Explain the following types of supply air outlets? [6]
- i) Grille outlets
 - ii) Ceiling diffuser outlets
 - iii) Slot diffuser outlets
- c) What is the function of “Ducts” in an air-conditioning system? Explain either “Radial Perimeter duct system” or “Extended plenum duct system” with neat sketch? [6]

